

JEE MAIN REVISION MASTER: RAY OPTICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 38 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Thin Lenses, Lens Maker's Formula & Optical Instruments	19 PYQs	Lenses & Instruments
High	Refraction, Snell's Law & Total Internal Reflection (TIR)	13 PYQs	Geometric Optics
Moderate	Prism Dispersion & Angle of Minimum Deviation	7 PYQs	Prisms

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Thin Lenses, Lens Maker's Formula & Optical Instruments	$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$, $\Delta P = P_1 + P_2$, $m_{\text{telescope}} = \frac{f_o}{f_e}$, $L = f_o + f_e$
Refraction, Snell's Law & Total Internal Reflection (TIR)	$\sin C = \frac{1}{n}$, $d_{\text{apparent}} = \frac{d_{\text{real}}}{n}$, $\Delta \text{Shift} = d\left(1 - \frac{1}{n}\right)$
Prism Dispersion & Angle of Minimum Deviation	$n = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$, $i = e = \frac{A + D_m}{2}$, $r = \frac{A}{2}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Thin Lenses, Lens Maker's Formula & Optical Instruments (Very High Yield — 19 PYQs)

Core Principle: Lens Maker's formula: $(1/f) = (n-1)\left((1/R_1) - (1/R_2)\right)$. Power $P = (1/f(m))$ in Dioptres.

Combination of lenses in contact: $P = P_1 + P_2 \implies (1/F) = (1/f_1) + (1/f_2)$. Telescope magnifying power in normal adjustment: $m = -(f_o / f_e)$, tube length $L = f_o + f_e$. Compound microscope: $m = -(v_o / u_o)\left(1 + (D / f_e)\right)$.

■ **Caution / JEE Pitfall:** When a glass lens ($n=1.5$) is immersed in water ($n=4/3$), its focal length increases by 4 times ($f_{(water)} = 4 f_{(air)}$) and its power decreases to $1/4$!

■ **Speed Trick:** If an equiconvex lens of focal length f is cut vertically in half, each plano-convex half has focal length $2f$. If cut horizontally in half, each piece retains focal length f .

Representative Source PYQs:

- [2025 Shift] Q#3 (MCQ): "A convex lens has power P . It is cut into two halves along its principal axis. Further one piece (out of two halves) is cut into two halves ..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "The image formed by an objective lens of a compound microscope is..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "An equiconvex lens of radius of curvature 14 cm is made up of two different materials. Left half and right half of vertical portion is made ..." [Ans: N/A]

3.2 Refraction, Snell's Law & Total Internal Reflection (TIR) (High Yield — 13 PYQs)

Core Principle: Snell's Law: $n_1 \sin i = n_2 \sin r$. Total Internal Reflection (TIR) occurs when light travels from a denser to a rarer medium at an angle of incidence exceeding the critical angle C : $\sin C = \frac{n_{(rarer)}}{n_{(denser)}}$. Applications: optical fibers, mirage, diamond brilliance. Apparent depth of a submerged object: $d' = d/n$.

■ **Caution / JEE Pitfall:** TIR can ONLY occur when light travels from an optically DENSER medium to an optically RARER medium, never vice versa!

■ **Speed Trick:** Critical angle for glass ($n=1.5$) is $C \approx 42^\circ$; for water ($n=4/3$) is $C \approx 49^\circ$; for diamond ($n=2.42$) is $C \approx 24^\circ$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "The critical angle for a monochromatic light going from medium A to medium B is θ . If the speed of light in medium A is V , then $t \dots$ " [Ans: C]
- [2026 Shift] Q#3 (MCQ): "The incorrect statement about refractive index for a pair of media is..." [Ans: N/A]
- [2026 Shift] Q#4 (MCQ): "The direction of a ray of light incident on a concave mirror is shown by PQ, while direction in which the ray would travel after reflection ..." [Ans: N/A]

3.3 Prism Dispersion & Angle of Minimum Deviation (Moderate Yield — 7 PYQs)

Core Principle: Prism formulas: $A = r_1 + r_2$ and $i + e = A + D$. At minimum deviation $D = D_m$: $i = e$ and $r_1 = r_2 = A/2$. Refractive index $n = (\sin((A+D_m)/2) / \sin(A/2))$. For a thin prism: $D = (n-1)A$.

■ **Caution / JEE Pitfall:** At minimum deviation in an equilateral prism ($A = 60^\circ$), the refracted ray inside the prism travels PARALLEL to the base.

■ **Speed Trick:** If $D_m = A$ for an equilateral prism, $n = (\sin A / \sin(A/2)) = 2\cos(A/2) = 2\cos 30^\circ = \sqrt{3} \approx 1.732$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "From the graph of angle of deviation versus angle of incidence for an equilateral prism, the refractive index of material of prism is..." [Ans: N/A]
- [2026 Shift] Q#1 (MCQ): "With reference to the figure shown below, match the following: List-I List-II (a) Angle of reflection (i) 60° (b) Value of $\alpha \dots$ " [Ans: A]
- [2025 Shift] Q#1 (MCQ): "If r and r^1 denotes the angles inside the prism having angle of prism 50° considering that during interval of time from $t=0 \dots$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

From the graph of angle of deviation versus angle of incidence for an equilateral prism, the refractive index of material of prism is

- (A) $\frac{\sqrt{3}}{2}$
- (B) $(3/2)$
- (C) $\sqrt{3}$
- (D) $\sqrt{2}$

Correct Answer: None

Solution Analysis: For an equilateral prism, the angle of prism is $A=60^\circ$. From the graph of angle of deviation δ versus angle of incidence i , the lowest value of deviation is the minimum deviation. Here, the graph shows $\delta_{(m)}=60^\circ$. The refractive index μ of the prism material is given by $\mu=\frac{\sin(\frac{A+\delta_{(m)}}{2})}{\sin(\frac{A}{2})}$. Now put $A=60^\circ$ and $\delta_{(m)}=60^\circ$ in the formula. $\mu=\frac{\sin(\frac{60^\circ+60^\circ}{2})}{\sin(\frac{60^\circ}{2})}$ $\mu=\frac{\sin(60^\circ)}{\sin(30^\circ)}$ Using $\sin 60^\circ=\frac{\sqrt{3}}{2}$ and

Benchmark Problem 2: KCET 2026 (Q#2)

The critical angle for a monochromatic light going from medium A to medium B is θ . If the speed of light in medium A is V , then the speed of light in medium B is

- (A) $V(1-\cos\theta)$
- (B) $(V / \cos\theta)$
- (C) $(V / \sin\theta)$ **[Correct]**
- (D) $V(1-\sin\theta)$

Correct Answer: C

Solution Analysis: For critical angle, light must be travelling from a denser medium to a rarer medium. Here, light goes from medium A to medium B, and the critical angle is θ . At the critical angle, the angle of refraction in medium B becomes 90° . Using Snell's law, $n_{(A)}\sin\theta=n_{(B)}\sin 90^\circ$. Since $\sin 90^\circ=1$, $n_{(A)}\sin\theta=n_{(B)}$. Now, refractive index is related to speed of light by $n=(c/v)$. So, $n_{(A)}=(c/V)$. Let the speed of light in medium B be $v_{(B)}$. Then, $n_{(B)}=\frac{c}{v_{(B)}}$. Substituting in the equation, $(c/V)\sin\theta=\frac{c}{v_{(B)}}$. Cancel

Benchmark Problem 3: KCET 2025 (Q#4)

The image formed by an objective lens of a compound microscope is

- (A) Real and diminished
- (B) Real and enlarged
- (C) Virtual and enlarged
- (D) Virtual and diminished

Correct Answer: None

Solution Analysis: The objective lens of a compound microscope produces a real, inverted and magnified image of the object. So the correct choice is: Option B: Real and enlarged